



GeoEnergy Space Nexus Research Institute

Three Pillars of Strategic Projects

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This document outlines the core project pillars that define the strategic direction of the GeoEnergy Space Nexus Research Institute. It is formatted for inclusion in the organization's GitHub repository as part of its public documentation, in compliance with GitHub's community and content guidelines.



Introduction

The GeoEnergy Space Nexus Research Institute (GESNRI) is a research-driven organization dedicated to advancing the intersection of geospatial science, space technology, energy sustainability, and humanitarian innovation. Our work is organized around three foundational pillars, each encompassing a set of interconnected project areas. Together, these pillars guide our research priorities, partnerships, and open-source contributions.

Each pillar below includes its scope, objectives, key activities, and intended outcomes. This structure is designed to align with collaborative, transparent, and reproducible research practices suitable for hosting on GitHub.



Pillar I

Space Data & Spatial Intelligence

1.1 Space Data Source

This project focuses on aggregating, curating, and distributing open space-based datasets including satellite imagery, remote sensing products, orbital telemetry, and earth observation archives. The aim is to build accessible, version-controlled repositories of space data that researchers and developers can use for analysis and application development.

Objectives

- Establish standardized pipelines for ingesting open satellite and earth observation data.
- Build GitHub-hosted catalogs and metadata indexes for discoverability.
- Provide reproducible scripts and notebooks for processing raw space data.
- Support interoperability with common geospatial data formats (GeoTIFF, NetCDF, STAC).

Key Activities

- Partnering with open data providers (e.g., space agencies, research consortia).
- Developing automated data update workflows using GitHub Actions.
- Publishing documentation and tutorials for accessing and using the datasets.

1.2 Spatial Data Collection

This project establishes systematic methods for collecting, validating, and organizing ground-based and aerial spatial data, including field surveys, GPS tracking, drone imagery, and crowdsourced geographic information. It complements the Space Data Source project by providing ground-truth and localized datasets.

Objectives

- Develop open protocols for spatial data collection in field research.
- Maintain version-controlled datasets with clear provenance and licensing.
- Enable integration of community-contributed geographic data.
- Provide quality assurance frameworks for spatial dataset validation.

Key Activities

- Designing standardized data collection templates and tools.
- Building GitHub repositories for dataset versioning and contribution tracking.
- Training partner organizations and volunteers on data collection best practices.



Pillar II

Climate, Energy & Humanitarian Impact

2.1 Humanitarian Data Source

This project develops open repositories of humanitarian data, including population displacement, disaster impact assessments, food security indicators, and infrastructure vulnerability maps. The goal is to support evidence-based humanitarian response through accessible, well-documented datasets.

Objectives

- Aggregate publicly available humanitarian datasets into unified, documented repositories.
- Ensure ethical data handling, anonymization, and privacy compliance.
- Provide tools for visualizing humanitarian indicators alongside spatial and climate data.
- Collaborate with humanitarian agencies to align with established data standards (e.g., HXL).

Key Activities

- Building open-source dashboards for humanitarian data exploration.
- Establishing data-sharing agreements with humanitarian organizations.
- Publishing case studies linking humanitarian needs with spatial analysis.

2.2 Smart-Climate Integration

This project integrates climate data streams, predictive modeling, and smart sensor networks to support climate monitoring, early warning systems, and adaptive planning. It bridges space data, ground sensors, and AI-driven analytics into unified climate intelligence tools.

Objectives

- Develop integrated climate monitoring dashboards combining satellite and IoT sensor data.
- Build predictive models for climate risk assessment using open-source machine learning tools.
- Create alerting systems for extreme weather and environmental hazards.
- Promote open standards for climate data exchange and interoperability.

Key Activities

- Developing open-source climate analytics libraries hosted on GitHub.
- Piloting smart-sensor deployments in partnership with local communities.

- Publishing research findings and model code for community review and reuse.

2.3 Sustainable Energy Development

This project focuses on research and tools that support the planning, modeling, and deployment of renewable and sustainable energy systems, with a focus on geospatial site selection, resource assessment, and energy access planning for underserved regions.

Objectives

- Develop geospatial models for renewable energy resource assessment (solar, wind, geothermal).
- Provide open-source tools for energy infrastructure planning and impact analysis.
- Support community energy access initiatives with data-driven site recommendations.
- Encourage open collaboration on sustainable energy datasets and methodologies.

Key Activities

- Building renewable resource maps using open satellite and climate datasets.
- Developing open-source planning toolkits for energy access projects.
- Partnering with local stakeholders on pilot sustainable energy projects.



Pillar III

Space Exploration & Innovation

3.1 Space Exploration

This project supports research, education, and open collaboration in space exploration topics, including mission concept studies, orbital mechanics simulations, planetary science datasets, and citizen-science space exploration initiatives.

Objectives

- Develop open-source simulation tools for orbital mechanics and mission planning.
- Curate educational resources and datasets related to planetary science and astronomy.
- Foster citizen-science participation in space exploration research.
- Collaborate with academic and space-agency partners on exploration-related research.

Key Activities

- Publishing open-source mission simulation and visualization tools.
- Hosting community challenges and hackathons focused on space exploration problems.
- Producing educational content and tutorials for students and researchers.

Cross-Pillar Synergies

While each project area has a distinct focus, the three pillars are designed to be mutually reinforcing. Space data underpins climate and energy modeling; spatial data collection validates and refines space-derived datasets; humanitarian data benefits from both spatial and climate intelligence; and space exploration research drives innovation that feeds back into earth observation and data collection capabilities.

GitHub Documentation Compliance

All project repositories under these pillars are maintained with clear licensing, contribution guidelines, codes of conduct, and documentation in accordance with GitHub's community guidelines and acceptable use policies. Each repository includes a README, LICENSE file, and CONTRIBUTING guide to ensure transparency and ease of collaboration.

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This document supports GitHub repository documentation policies and may be published under the organization's open documentation guidelines.